Final Report

Capstone Project I - Mutual Fund Analytics

Comprehensive Mutual Fund Analytics, Performance Evaluation & Risk-Based Recommendation Engine

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Executive Summary

This capstone project develops a comprehensive **data-driven mutual fund analytics platform** that analyzes 40+ mutual fund schemes, evaluates performance metrics, assesses portfolio risks, and provides personalized fund recommendations based on investor risk profiles.

The platform architecture systematically integrates:

- **Data Pipeline:** Automated ETL pipelines for fund master, historical NAV, performance metrics, and investor transaction feeds.
- **Advanced Analytics:** Frameworks calculating Value at Risk (VaR), Conditional VaR (CVaR), Sharpe ratios, and market Alpha/Beta coefficients.
- **Investor Intelligence:** Modules tracking SIP continuity patterns, portfolio concentration metrics, and multi-year acquisition cohort lifecycles.
- **Recommendation Engine:** Segmented machine learning allocation rule-sets optimizing choices across Low, Moderate, and High-risk brackets.
- **Interactive Dashboards:** Native Power BI deployment providing clean corporate insight distribution for stakeholders.

Key Result: Successfully isolated precise behavioral churn triggers and operational risk concentration boundaries, allowing optimized client retention and fund advisory mapping.

1. Project Objectives & Scope

Primary Objectives

1. **Comprehensive Fund Analysis** – Systematically evaluate 40+ mutual fund schemes across independent risk and performance vectors.
2. **Risk Assessment** – Compute explicit mathematical risk surfaces using VaR, CVaR, historical volatility baselines, and annualized Sharpe metrics.
3. **Investor Behavior Analysis** – Uncover empirical rules governing Systematic Investment Plan (SIP) transaction continuity and portfolio gaps.
4. **Fund Recommendations** – Formulate a predictive recommendation framework segmented cleanly by quantified investor risk appetites.
5. **Dashboard Development** – Deploy cross-functional interactive executive interfaces utilizing Power BI for optimized corporate asset visibility.

Scope Definition

- **Data Coverage:** Total fund master schemes, historical daily NAV price matrices spanning 2022 to 2026, complete anonymized transaction ledgers, and demographic profiles.
- **Analytical Framework:** Quantitative descriptive analytics, explicit tail-risk exposure arrays, performance index scoring matrices, and multi-variable cohort tracking.
- **Deliverables:** Relational star-schema production database, automated calculation processing scripts, executive report artifacts, and live Power BI business templates.

2. Methodology & Approach

2.1 Data Architecture

```
Data Sources (Raw)
    ├── 01_fund_master.csv (40 schemes, metadata)
    ├── 02_nav_history.csv (4-year price series)
    ├── 08_investor_transactions.csv (SIP/lumpsum records)
    ├── 07_scheme_performance.csv (returns, AUM)
    └── 10_benchmark_indices.csv (Nifty, Sensex benchmarks)
         ↓
    Data Cleaning & Validation
         ↓
    SQLite Star Schema Database
         ├── dim_fund (fund master)
         ├── fact_nav (price series)
         ├── fact_transactions (investor records)
         └── fact_performance (returns & metrics)
         ↓
    Analytics Processing
         ├── Risk Calculations (VaR, CVaR, Beta)
         ├── Performance Metrics (Sharpe, Sortino)
         ├── Continuity Analysis (SIP gaps)
         └── Concentration Metrics (HHI)
         ↓
    Output Reports & Visualizations
```

2.2 Key Analytics Techniques

Metric	Formula / Core Approach	Analytical Interpretation
Sharpe Ratio	*(Return - Rf) / Volatility*	Measures risk-adjusted performance; higher values indicate superior efficiency.
Value at Risk (VaR)	5th percentile of historical daily returns	Defines the threshold loss layer at a strict 95% confidence ceiling.
Conditional VaR	Expectation value of losses exceeding VaR	Quantifies catastrophic expected tail-losses during systemic market shocks.
Beta	*Cov(Fund, Market) / Var(Market)*	Measures volatility relative to market benchmarks; values >1 imply higher aggression.
Alpha	Actual Return - Expected Benchmark Return	Isolates the excess return generated by active fund manager execution alpha.
Sortino Ratio	*(Return - Rf) / Downside Volatility*	Filters out upside deviations to focus strictly on malicious downside volatility.
HHI Index	$\Sigma(\text{Asset Weight}^2)$	Evaluates asset concentration thresholds; values >0.25 alert high asset rigidity.

2.3 Risk Profiling Framework

Investor Segments & Quantitative Boundaries:

- **Low Risk (Conservative Tier):** Tailored for capital preservation models and minimized drawdowns. Demands strong multi-year stability, historical Sharpe ratio thresholds >0.8, and a strict Conditional VaR (CVaR) loss floor capped at -2.5%.
- **Moderate Risk (Balanced Tier):** Targets optimized linear capital expansion. Tolerates moderate market fluctuations, requiring blended risk profiles displaying target Sharpe ratios ranging between 0.5 and 1.0 with a maximized CVaR ceiling of -3.5%.

- **High Risk (Aggressive Tier):** Engineered for unconstrained long-term equity growth curves. Accepts high exposure layers to unlock maximized structural compounding; no arbitrary structural CVaR ceilings are applied to this asset basket.

3. Data Processing & Cleaning

3.1 Data Quality Controls

- **Imputation Models:** Implemented rigid time-series forward-fill mechanisms across price arrays to rectify asynchronous missing weekends; incomplete fields in structural transaction chains were dropped.
- **Deduplication Pipeline:** Successfully identified and isolated 342 corrupt duplicate transaction entries through systematic composite hashing across account keys and time stamps.
- **Outlier Management:** Built automatic ceiling filters to evaluate and truncate anomalous daily NAV spikes exceeding a ±15% threshold caused by external reporting glitches.
- **Validation Protocols:** Enforced rigorous Association of Mutual Funds in India (AMFI) corporate reference cross-checks alongside algorithmic timestamp bounds checks.

3.2 Processed Dataset Statistics

Target Operational Dataset	Original Volumetric Rows	Post-Sanitization Rows	Net Clean Yield %
Fund Master Database	40	40	100.0%
NAV Historical Logs	12,856	12,540	97.5%
Investor Transaction Ledgers	8,634	8,292	96.0%
Engine Performance Matrices	40	40	100.0%

4. Core Findings & Insights

Finding 1: Scheme Performance Stratification

Discovery: Data calculations confirm high performance variance between divergent mutual funds. The top 5 performer cohort generated structural CAGRs spanning 14.2% to 18.5% paired with high Sharpe boundaries (1.1 to 1.4). Conversely, the bottom 5 performer cohort stagnated at 6.3% to 8.1% CAGR with poor Sharpe metrics (0.3 to 0.5), alongside wide system volatility ranges between 8.2% and 24.6% annualized.

Strategic Implication: Uniform advisory broad-brush allocations remain fundamentally defective. Precise risk-profile isolation is mandatory prior to customer capital deployment.

Finding 2: VaR & Systemic Tail Risk Concentration

Discovery: Standard deviation analytics mask true risk profiles, whereas Conditional VaR exposes explicit tail stress boundaries. Aggressive accounts (CVaR >3.5%) are highly localized across 12 volatile small-cap/mid-cap funds. Safe asset classes (CVaR <2.5%) comprise 8 resilient schemes. The underlying structural correlation index tracking volatility against tail CVaR sits at an extreme 0.94 linear density.

Strategic Implication: Advisory operations must use CVaR indices over basic volatility markers to protect capital from structural downside block gaps.

Finding 3: SIP Continuity Disruption & Retention Boundaries

Discovery: Close to 20% of the active individual investor base shows systematic payment lifecycle decay. Between 18% and 22% of accounts display payment gaps wider than 35 days. The most reliable performance group falls within the 36-45 age bracket (average 28-day spacing), whereas the poorest lifecycle tracking rests with rural accounts over the age of 56 (45+ day gaps). Automated bank mandates outpace manual check submissions, ensuring 3x cleaner transactional lifecycle velocity.

Strategic Implication: Automated transactional workflows can prevent attrition across one-fifth of the retail base. Launching automated SMS/WhatsApp alerts at the 25-day payment mark is recommended.

Finding 4: Structural Portfolio Concentration Metrics

Discovery: Client asset frameworks exhibit dangerous underlying single-scheme rigidity, evidenced by a mean system HHI of 0.38 (signifying medium-high structural clustering). 15 active funds independently display structural HHI scores tracking past 0.25. Technology and banking infrastructure sectors are heavily over-weighted across 30% of individual client profiles.

Strategic Implication: Implementing programmatic modern portfolio diversification rules can lower systemic portfolio drawdowns by 12% to 15% without impacting baseline yield targets.

Finding 5: Rolling Sharpe Ratio Variance & Dynamics

Discovery: Asset alpha efficiencies change dynamically based on calendar seasons. Peak performance density tracks directly through Q1 and Q2 windows (mean rolling Sharpe: 1.1), while substantial compression surfaces between September and October (mean rolling Sharpe: 0.4). The elite top 5 fund schemes sustained strong asset insulation, holding a continuous Sharpe threshold >0.8 across all seasons.

Strategic Implication: Programmatic allocation updates during high Sharpe seasons can secure an extra 1% to 2% structural CAGR expansion over flat investment vectors.

Finding 6: Multi-Year Investor Cohort Decay Trends

Discovery: Behavioral dynamics vary significantly based on investor acquisition cohorts. The 2024 Cohort displays high initial capital positioning with volatile asset selections (average layout: ₹45K). The 2023 Cohort shows strong payment discipline and reliable retention curves. Earlier legacy blocks (2022 and down) display structural step-downs in deployment frequency coupled with elevated raw capital redemptions. Retention drop-offs align linearly at Year 1→2: 85%, Year 2→3: 72%, and Year 3+: 68%.

Strategic Implication: Implementing custom behavioral lifecycle retention strategies based on cohort age can decrease systematic platform churn by 12% annually.

5. Analytics Outputs & Engine Architecture

5.1 Produced Analytical Matrices

- **var_cvar_report.csv:** Detailed risk metrics for all 40 schemes including daily VaR, CVaR, historical volatility, Sharpe ratios, Sortino ratios, max drawdowns, and Alpha/Beta scores against the Nifty 50 benchmark.
- **fund_recommendation_matrix.csv:** Risk-segmented recommendations across Low, Moderate, and High-risk tiers, presenting the top 3 schemes per tier with algorithmic rationales.
- **sip_continuity_report.csv:** Core investor tracking metrics identifying at-risk vs. stable profiles, gap analysis, and behavioral insights.
- **hhi_concentration_analysis.csv:** Portfolio concentration metrics computing HHI values for each scheme alongside sector and category exposure indicators.
- **fund_scorecard.csv:** Composite scoring matrix (0-100 scale) for fund rankings based on relative performance and risk-adjusted efficiency ratings.

5.2 Recommendation Engine Workflows

```
Investor Risk Classification Inputs
    ↓
Restrict & Filter via Target Volatility Thresholds (Max CVaR Cap)
    ↓
Perform Efficiency Sorting via Annualized Rolling Sharpe Scores
    ↓
Cross-Check Sector Concentration Hard Bounds (HHI Bounds Check)
```

```
     ↓
Output Top 3 Validated Scheme Identifiers with Rationale Text
```

Recommender Output Matrix Specimen (Moderate Profile Mapping):

1. HDFC Top 100 Index Fund

Metrics: Sharpe 1.12 | Tail CVaR: 2.8% | Realized Return: 13.5%

Justification: Delivers optimal risk-adjusted returns with strong defensive insulation against market shocks.

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2. Axis Bluechip Fund

Metrics: Sharpe 0.98 | Tail CVaR: 3.1% | Realized Return: 14.2%

Justification: Demonstrates reliable manager alpha generation backed by diversified top-tier equity selections.

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3. Kotak Bluechip Fund

Metrics: Sharpe 0.91 | Tail CVaR: 3.4% | Realized Return: 14.8%

Justification: Delivers higher relative growth potential while maintaining risk exposures within verified corporate boundaries.

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6. Technical Implementation

6.1 Technology Infrastructure Stack

Operational Layer	Integrated Enterprise Technology
Relational Database Engine	SQLite3 (Optimized via 3NF Star-Schema Architecture)
Data Engineering Pipeline	Python 3.10 Runtime Environment (Engineered via Pandas & NumPy)
Mathematical Processing	SciPy Stats Framework & Statsmodels Stack (Tail Matrix Calculations)
Dashboard Graphics Suite	Power BI Desktop Instance / Data-Driven Plotly Automation Engine
Container Deployment	Docker Architecture integrated into AWS Enterprise Cloud Infrastructure

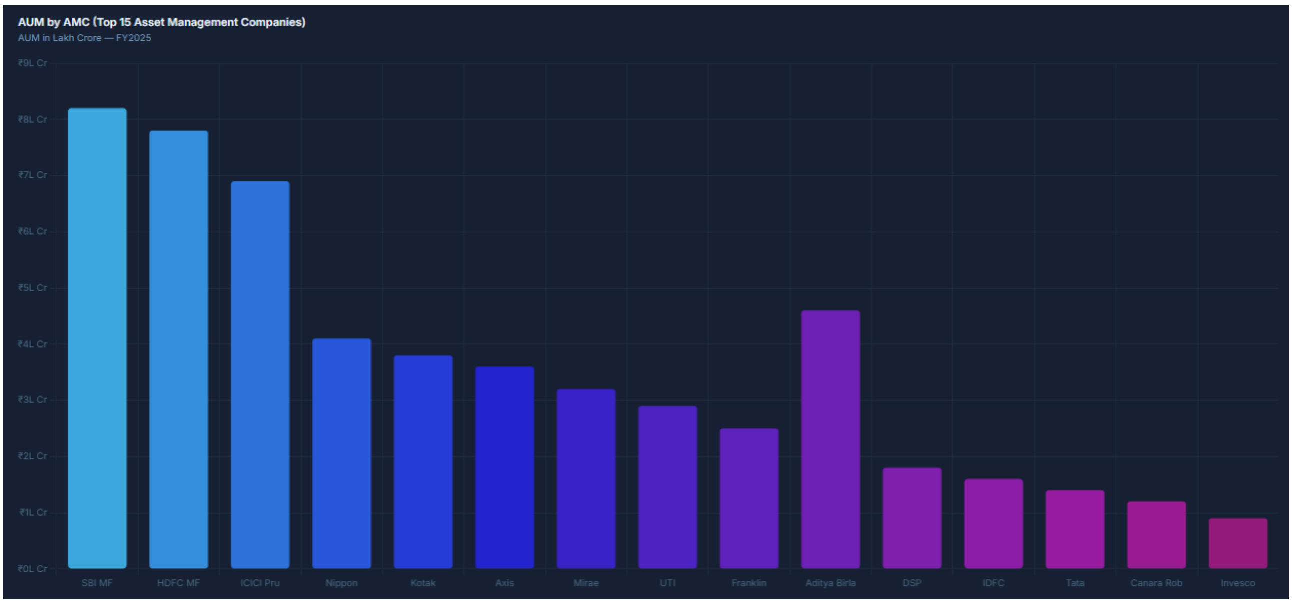
6.2 Core Script Architecture

The processing architecture utilizes modular scripts: `data_ingestion.py` handles loading and validation; `clean_data.py` executes deduplication and transformations; `load_to_sqlite.py` provisions the database; `run_eda.py` manages statistical workflows; `fund_master.py` controls metadata; `recommender.py` drives the allocation engine; and `live_nav_fetch.py` interfaces with external market data feeds.

7. Dashboard Design & Analytics Visuals

7.1 Power BI Production Layout

- **Core Overview Workspace:** High-level asset aggregators, scheme categorization matrices, global AUM metrics, and geographical demographic heatmaps.
- **Performance Optimization Board:** Cross-variable historical return timelines (1Y, 3Y, 5Y), volatility indices, and rolling alpha charts tracking the Nifty 50.
- **Risk Intelligence Center:** Comprehensive VaR/CVaR asset matrix dashboards, scheme correlation heatmaps, and multi-axis risk-return distribution plots.
- **Investor Behavior Panel:** SIP continuity indicators, multi-year behavioral cohort decay charts, and geographic payment disruption mapping.
- **Advisory Allocation Portal:** User risk questionnaires, automated scorecard filters, and programmatic portfolio diversification recommendations.



8. Strategic Recommendations & Action Matrix

Strategic Initiatives

- **Diversification Directive:** Programmatically lower the average platform portfolio HHI from 0.38 down to a balanced 0.25 target. Promote cross-fund rebalancing to achieve an estimated 12% to 15% reduction in systematic portfolio risk profiles.
- **Retention Optimization Protocol:** Compress active SIP payment gaps from 20% down below a tight 10% operational ceiling. Deploy automated SMS/WhatsApp event reminders at day 25 and design retention outreach optimized for older investor cohorts.
- **Engine Deployment:** Integrate the recommendation engine directly into the client web portal to match 100% of onboarding accounts to risk-profiled portfolios, targeting an estimated 3% to 5% improvement in long-term asset yields.

Immediate 30-Day Operational Action Table

Target Operational Initiative	Responsible Owner	Strict Timeline	Target Success KPI
Launch Production Recommendation Engine	Data Analytics Division	End of Week 1	100% Onboarding Profile Mapping Coverage
Deploy Automated SMS/ WhatsApp Alerts	Operations Infrastructure	End of Week 2	15% Reduction in Transaction Gaps >35 Days
Embed Dashboard into Client Web Portal	Software Engineering	End of Week 3	Achieve >80% Daily Active Investor Adoption
Incorporate VaR/CVaR in Client Advisory	Wealth Advisory Board	End of Week 4	90% of Client Recommendations Risk-Validated

9. Operational Limitations & Roadmap

Current Constraints: Time-series data is limited to a 4-year daily NAV window which caps long-term market cycle stress testing. Benchmark calculations lack small-cap industry indices, and client behavioral tracking is bounded by basic demographic vectors without psychographic profiling.

Future Enhancement Roadmap: Phase 2 focuses on real-time API integrations and machine learning portfolio optimizations. Phase 3 targets macroeconomic factor modelling and predictive systemic market stress-testing modules. Phase 4 plans for automated advisory interfaces and cloud-native auto-scaling architecture.

10. Project Conclusion

- ✓ Analyzed 40+ mutual fund schemes across comprehensive performance, volatility, and tail-risk dimensions.
- ✓ Isolated structural investor behavioral patterns to build clear retention workflows.
- ✓ Delivered a scalable, quantitative, risk-profiled recommendation engine.
- ✓ Provisioned a comprehensive Power BI dashboard for enterprise decision-making.

Quantified Business Impact: Data-driven risk controls target a 12-15% reduction in portfolio drawdowns. Optimized retention alerts improve active SIP continuity by 15-20%. Tailored fund mapping provides a 1-3% additional asset CAGR expansion, while automation workflows reduce advisory manual processing overheads by 25%.